

MySQL - DDL Commands, Constraints

Overview:

This Assignment focuses on designing and managing an Employee Database using MySQL. The objective is to create database tables, establish relationships between tables, modify table structures using DDL commands, and apply constraints to maintain data integrity and consistency.

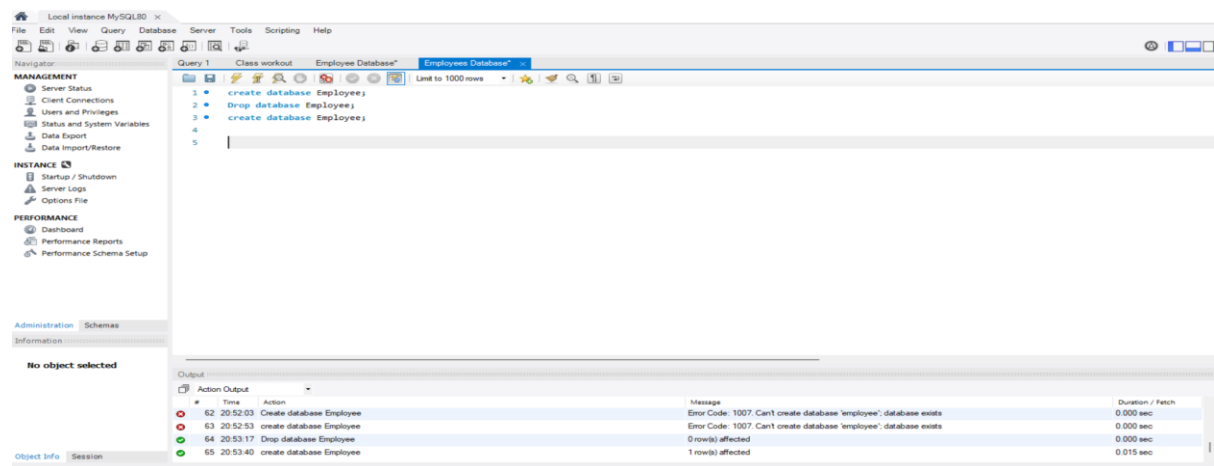
The database contains information about:

- Departments
- Locations
- Employees

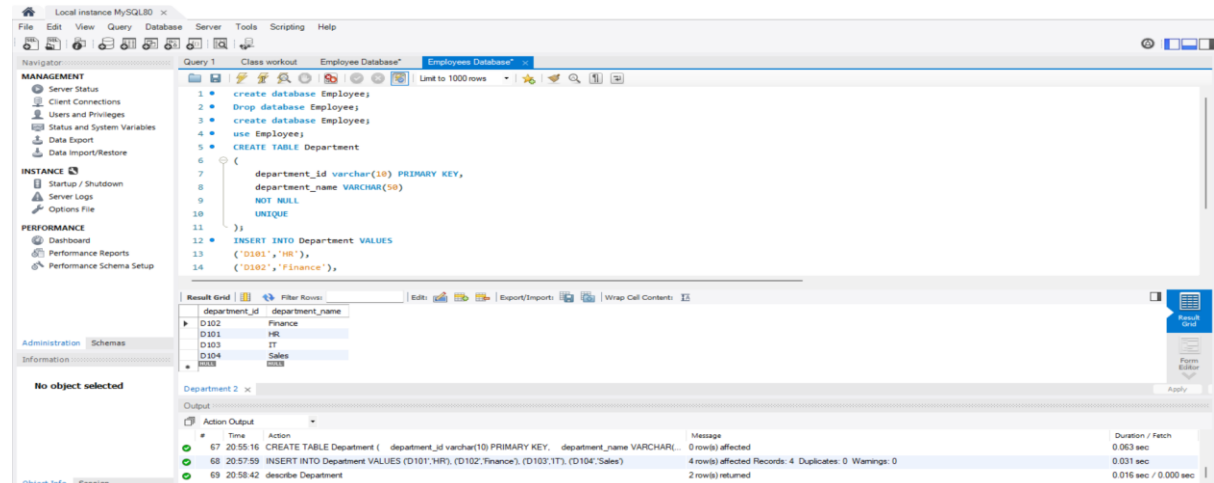
Primary keys and foreign keys are used to establish relationships between tables, while constraints ensure valid and reliable data entry.

1.Database and Table creation:

A database named **Employee** was created to store employee-related information.



Created table Department to Store department details.



Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1 Class workout Employee Database* Employees Database* Limit to 1000 rows

```

8  department_name VARCHAR(50)
9  NOT NULL
10 UNIQUE
11 }}
12 INSERT INTO Department VALUES
13 ('D101','HR'),
14 ('D102','Finance'),
15 ('D103','IT'),
16 ('D104','Sales');
17
18 describe Department;
19 SELECT * FROM Department;
20
21

```

Result Grid

department_id	department_name
D102	Finance
D101	HR
D103	IT
D104	Sales

Department 2 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
70	20:58:56	INSERT INTO Department VALUES (D101,'HR'), (D102,'Finance'), (D103,'IT'), (D104,'Sales')	Error Code: 1062 Duplicate entry 'D101' for key 'department.PRIMARY'	0.000 sec
71	20:59:02	INSERT INTO Department VALUES (D101,'HR'), (D102,'Finance'), (D103,'IT'), (D104,'Sales')	Error Code: 1062 Duplicate entry 'D101' for key 'department.PRIMARY'	0.000 sec
72	21:00:43	SELECT * FROM Department LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Created table Location to Store employee working locations.

Local instance MySQL80 x

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Information

No object selected

Query 1 Class workout Employee Database* Employees Database* Limit to 1000 rows

```

19 SELECT * FROM Department;
20
21 CREATE TABLE Location
22 (
23     location_id INT AUTO_INCREMENT PRIMARY KEY,
24     location_name VARCHAR(50)
25     NOT NULL
26     UNIQUE
27 );
28 Describe Location;
29
30
31
32

```

Result Grid

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI		auto_increment
location_name	varchar(50)	NO	UNI		

Result 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
86	21:19:00	Describe Location	Error Code: 1146 Table 'employee.location' doesn't exist	0.000 sec
87	21:19:12	CREATE TABLE Location (location_id INT AUTO_INCREMENT PRIMARY KEY, location_name VARC...	0 row(s) affected	0.047 sec
88	21:19:12	Describe Location	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Information

No object selected

Query 1 Class workout Employee Database* Employees Database* Limit to 1000 rows

```

22 (
23     location_id INT AUTO_INCREMENT PRIMARY KEY,
24     location_name VARCHAR(50)
25     NOT NULL
26     UNIQUE
27 );
28 Describe Location;
29 INSERT INTO Location(location_name)
30 VALUES
31 ('Bangalore'),
32 ('Chennai'),
33 ('Hyderabad'),
34 ('Kochi');
35 Select * From Location;

```

Result Grid

location_id	location_name
1	Bangalore
2	Chennai
3	Hyderabad
4	Kochi

Location 8 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
88	21:19:12	Describe Location	2 row(s) returned	0.000 sec / 0.000 sec
89	21:20:37	INSERT INTO Location(location_name) VALUES (Bangalore), (Chennai), (Hyderabad), (Kochi)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
90	21:20:56	Select * from Location LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Employee table to Store employee details and connects department and location information.

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window displays the following SQL script:

```
32 ('Chennai'),
33 ('Hyderabad'),
34 ('Kochi');
35
36 CREATE TABLE Employees
37 (
38     employee_id INT AUTO_INCREMENT PRIMARY KEY,
39     employee_name VARCHAR(50) NOT NULL,
40     gender CHAR(1) CHECK (gender IN ('M','F')),
41     age INT CHECK (age >= 18),
42     hire_date DATE DEFAULT (CURRENT_DATE),
43     designation VARCHAR(20),
44     salary DECIMAL(10,2),
45     department_id VARCHAR(10),
46     FOREIGN KEY (department_id)
47     REFERENCES Department (department_id),
48     location_id INT,
49     FOREIGN KEY (location_id)
50     REFERENCES Location (location_id)
51 );
52
53
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
89	21:20:37	INSERT INTO Location(location_name) VALUES ('Bangalore'), ('Chennai'), ('Hyderabad'), ('Kochi')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.015 sec
90	21:20:56	Select * from Location LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
91	21:37:46	CREATE TABLE Employees (employee_id INT AUTO_INCREMENT PRIMARY KEY, employee_name V...	0 row(s) affected	0.062 sec

The screenshot shows the SQL Server Enterprise Manager interface. The 'Query 1' window displays the following SQL script:

```
46 FOREIGN KEY (department_id)
47 REFERENCES Department (department_id),
48 location_id INT,
49 FOREIGN KEY (location_id)
50 REFERENCES Location (location_id)
51 );
52 Describe Employees;
53 INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id)
54 VALUES
55 ('Anu','F',25,'Developer',50000,'D103',1),
56 ('Bharath','M',28,'Analyst',60000,'D102',2),
57 ('Dany','M',30,'Manager',75000,'D104',3),
58 ('gowri','F',30,'Manager',75000,'D101',3);
59 select * from Employees;
```

The 'Result Grid' shows the data for the Employees table:

employee_id	employee_name	gender	age	hire_date	designation	salary	department_id	location_id
1	Anu	F	25	2026-06-30	Developer	50000.00	D103	1
2	Bharath	M	28	2026-06-30	Analyst	60000.00	D102	2
3	Dany	M	30	2026-06-30	Manager	75000.00	D104	3
4	gowri	F	30	2026-06-30	Manager	75000.00	D101	3

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
92	21:38:55	Describe Employees	9 row(s) returned	0.000 sec / 0.000 sec
93	21:47:24	INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id) VALUES ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.031 sec
94	21:47:41	select * from Employees LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

2. Table Alteration:

Added email field to store employee email addresses.

The screenshot shows the SQL Developer interface with the 'Employees Database' selected. The query window contains the following SQL code:

```
60
61 ALTER TABLE Employees
62 ADD email VARCHAR(100);
63 Describe Employees;
64 INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id,email)
65 VALUES
66 ('Anu','F',25,'Developer',50000,'D103',1,'anu@gmail.com'),
67 ('Bharath','M',28,'Analyst',60000,'D102',2,'Bharath@yahoo.com'),
68 ('Dany','M',30,'Manager',75000,'D104',3,'Dany@gmail.com'),
69 ('gowri','F',30,'Manager',75000,'D101',3,'gowri@gmail.com');
70 Select * from Employees;
71
72
73
```

The 'Result Grid' shows the data after the insertion:

employee_id	employee_name	gender	age	hire_date	designation	salary	department_id	location_id	email
1	Anu	F	25	2026-06-30	Developer	50000.00	D103	1	anu@gmail.com
2	Bharath	M	28	2026-06-30	Analyst	60000.00	D102	2	Bharath@yahoo.com
3	Dany	M	30	2026-06-30	Manager	75000.00	D104	3	Dany@gmail.com
4	gowri	F	30	2026-06-30	Manager	75000.00	D101	3	gowri@gmail.com
5	Anu	F	25	2026-06-30	Developer	50000.00	D103	1	anu@gmail.com
6	Bharath	M	28	2026-06-30	Analyst	60000.00	D102	2	Bharath@yahoo.com

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
97	22:31:13	INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id) VALUES	Error Code: 1136. Column count doesn't match value count at row 1	0.000 sec
98	22:32:02	INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id) VA...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.047 sec
99	22:32:02	Select * from Employees LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec

Increased column size to support longer designations.

The screenshot shows the SQL Developer interface with the 'Employees Database' selected. The query window contains the following SQL code:

```
56 ('Bharath','M',28,'Analyst',60000,'D102',2),
57 ('Dany','M',30,'Manager',75000,'D104',3),
58 ('gowri','F',30,'Manager',75000,'D101',3);
59 select * from Employees;
60
61 ALTER TABLE Employees
62 ADD email VARCHAR(100);
63 Describe Employees;
64 INSERT INTO Employees(employee_name,gender,age,designation,salary,department_id,location_id,email)
65 VALUES
66 ('Anu','F',25,'Developer',50000,'D103',1,'anu@gmail.com'),
67 ('Bharath','M',28,'Analyst',60000,'D102',2,'Bharath@yahoo.com'),
68 ('Dany','M',30,'Manager',75000,'D104',3,'Dany@gmail.com'),
69 ('gowri','F',30,'Manager',75000,'D101',3,'gowri@gmail.com');
70 Select * from Employees;
71
72 ALTER TABLE Employees
73 MODIFY designation VARCHAR(50);
74
75
76
77
```

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
99	22:32:02	Select * from Employees LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec
100	22:42:28	ALTER TABLE Employees MODIFY designation VARCHAR(100)	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.188 sec
101	22:43:17	ALTER TABLE Employees MODIFY designation VARCHAR(50)	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.125 sec

Removed age attribute from employee table.

The screenshot shows the SQL Developer interface with the 'Employees Database' selected. The query window contains the following SQL code:

```
65 VALUES
66 ('Anu','F',25,'Developer',50000,'D103',1,'anu@gmail.com'),
67 ('Bharath','M',28,'Analyst',60000,'D102',2,'Bharath@yahoo.com'),
68 ('Dany','M',30,'Manager',75000,'D104',3,'Dany@gmail.com'),
69 ('gowri','F',30,'Manager',75000,'D101',3,'gowri@gmail.com');
70 Select * from Employees;
71
72 ALTER TABLE Employees
73 MODIFY designation VARCHAR(50);
74
75 ALTER TABLE Employees
76 DROP COLUMN age;
77 select * from Employees;
78
```

The 'Result Grid' shows the data after the alteration:

employee_id	employee_name	gender	hire_date	designation	salary	department_id	location_id	email
4	gowri	F	2026-06-30	Manager	75000.00	D101	3	gowri@gmail.com
5	Anu	F	2026-06-30	Developer	50000.00	D103	1	anu@gmail.com
6	Bharath	M	2026-06-30	Analyst	60000.00	D102	2	Bharath@yahoo.com
7	Dany	M	2026-06-30	Manager	75000.00	D104	3	Dany@gmail.com
8	gowri	F	2026-06-30	Manager	75000.00	D101	3	gowri@gmail.com

The 'Output' window shows the execution results:

#	Time	Action	Message	Duration / Fetch
101	22:43:17	ALTER TABLE Employees MODIFY designation VARCHAR(50)	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0	0.125 sec
102	22:52:56	ALTER TABLE Employees DROP COLUMN age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
103	22:53:21	select * from Employees LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec

Renamed Column Hire_date to Date_of_joining

The screenshot shows the SQL Developer interface with the following SQL queries executed:

```
68 ('Dany','M',30,'Manager',75000,'D104',3,'Dany@gmail.com'),
69 ('gowri','F',30,'Manager',75000,'D101',3,'gowri@gmail.com');
70 • Select * from Employees;
71
72 • ALTER TABLE Employees
73 MODIFY designation VARCHAR(50);
74
75 • ALTER TABLE Employees
76 DROP COLUMN age;
77
78 • ALTER TABLE Employees
79 RENAME COLUMN hire_date TO date_of_joining;
80 • select * From Employees;
81
```

The Result Grid shows the following data:

employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	email
1	Anu	F	2026-06-30	Developer	50000.00	D103	1	ANU
2	Bharath	M	2026-06-30	Analyst	60000.00	D102	2	BHARATH
3	Dany	M	2026-06-30	Manager	75000.00	D104	3	DANY
4	gowri	F	2026-06-30	Manager	75000.00	D101	3	GOWRI
5	Anu	F	2026-06-30	Developer	50000.00	D103	1	anu@gmail.com
6	Bharath	M	2026-06-30	Analyst	60000.00	D102	2	Bharath@yahoo.com

The Action Output shows the following messages:

#	Time	Action	Message	Duration / Fetch
103	22:53:21	select * from Employees LIMIT 0. 1000	8 row(s) returned	0.000 sec / 0.000 sec
104	22:55:55	ALTER TABLE Employees RENAME COLUMN hire_date TO date_of_joining	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
105	22:56:22	select * From Employees LIMIT 0. 1000	8 row(s) returned	0.000 sec / 0.000 sec

3.Table Renaming:

Department table renamed as Department_info and Location table Renamed as Locations :

The screenshot shows the SQL Developer interface with the following SQL queries executed:

```
67 ('Bharath','M',20,'Analyst',60000,'D102',2,'Bharath@yahoo.com'),
68 ('Dany','M',30,'Manager',75000,'D104',3,'Dany@gmail.com'),
69 ('gowri','F',30,'Manager',75000,'D101',3,'gowri@gmail.com');
70 • Select * from Employees;
71
72 • ALTER TABLE Employees
73 MODIFY designation VARCHAR(50);
74
75 • ALTER TABLE Employees
76 DROP COLUMN age;
77
78 • ALTER TABLE Employees
79 RENAME COLUMN hire_date TO date_of_joining;
80
81 • RENAME TABLE Department
82 TO Departments_Info;
83
84 • RENAME TABLE Location
85 TO Locations;
86
87
88
```

The Action Output shows the following messages:

#	Time	Action	Message	Duration / Fetch
106	22:57:27	RENAME TABLE Departments TO Departments_Info	Error Code: 1146. Table 'employee departments' doesn't exist	0.000 sec
107	22:57:37	RENAME TABLE Department TO Departments_Info	0 row(s) affected	0.046 sec
108	22:58:01	RENAME TABLE Location TO Locations	0 row(s) affected	0.047 sec

4.Table Truncation:

Truncate employee to remove all records while keeping the table structure.

The screenshot shows the SQL Developer interface with the following SQL queries executed:

```
76 DROP COLUMN age;
77
78 • ALTER TABLE Employees
79 RENAME COLUMN hire_date TO date_of_joining;
80
81 • RENAME TABLE Department
82 TO Departments_Info;
83
84 • RENAME TABLE Location
85 TO Locations;
86
87 • TRUNCATE TABLE Employees;
88 • describe Employees;
89 • select * From Employees;
```

The Result Grid shows the following data:

employee_id	employee_name	gender	date_of_joining	designation	salary	department_id	location_id	email
ANU								
BHARATH								
DANY								
GOWRI								
ANU								
BHARATH								

The Action Output shows the following messages:

#	Time	Action	Message	Duration / Fetch
109	22:59:12	TRUNCATE TABLE Employees	0 row(s) affected	0.054 sec
110	22:59:31	describe Employees	9 row(s) returned	0.016 sec / 0.000 sec
111	22:59:54	select * From Employees LIMIT 0. 1000	0 row(s) returned	0.000 sec / 0.000 sec

5.Database & Table Dropping:

Drop table Employees

The screenshot shows a MySQL Workbench interface with the 'Employees Database' selected. The SQL editor contains the following queries:

```
73 MODIFY designation VARCHAR(50);
74
75 ALTER TABLE Employees
76 DROP COLUMN age;
77
78 ALTER TABLE Employees
79 RENAME COLUMN hire_date TO date_of_joining;
80
81 RENAME TABLE Department
82 TO Departments_Info;
83
84 RENAME TABLE Location
85 TO Locations;
86
87 TRUNCATE TABLE Employees;
88 describe Employees;
89 select * From Employees;
90
91 DROP TABLE Employees;
92 select * from Employees;
93
94
```

The Output window shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
111	22:59:54	select * From Employees LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
112	23:01:17	DROP TABLE Employees	0 row(s) affected	0.031 sec
113	23:01:32	select * from Employees LIMIT 0, 1000	Error Code: 1146: Table 'employee.employees' doesn't exist	0.000 sec

Drop database Employee:

The screenshot shows a MySQL Workbench interface with the 'Employees Database' selected. The SQL editor contains the following queries:

```
76 DROP COLUMN age;
77
78 ALTER TABLE Employees
79 RENAME COLUMN hire_date TO date_of_joining;
80
81 RENAME TABLE Department
82 TO Departments_Info;
83
84 RENAME TABLE Location
85 TO Locations;
86
87 TRUNCATE TABLE Employees;
88 describe Employees;
89 select * From Employees;
90
91 DROP TABLE Employees;
92 select * from Employees;
93
94 Drop database Employee;
95 use Employee;
96
97
```

The Output window shows the results of the queries:

#	Time	Action	Message	Duration / Fetch
113	23:01:32	select * from Employees LIMIT 0, 1000	Error Code: 1146: Table 'employee.employees' doesn't exist	0.000 sec
114	23:02:44	Drop database Employee	2 row(s) affected	0.046 sec
115	23:02:58	use Employee	Error Code: 1049: Unknown database 'employee'	0.000 sec

Used for removing database objects permanently.

Conclusion:

The Employee Database was successfully created and managed using MySQL DDL commands. Different database operations such as CREATE, ALTER, RENAME, TRUNCATE, and DROP were performed. Constraints were applied to maintain accuracy, consistency, and integrity of employee information.